Exercises

Exercise 2.1 · Modulus and argument. Determine the modulus and an argument of: $1-i$; $-\sqrt{3}+i$; -5; $4i$.

Exercise 2.2 · From the trigonometric form to the algebraic form. Write in algebraic form: $2\left(\cos\frac{2\pi}{3}+i\sin\frac{2\pi}{3}\right)$; $\sqrt{2}\left(\cos\left(-\frac{\pi}{4}\right)+i\sin\left(-\frac{\pi}{4}\right)\right)$; $6\left(\cos\frac{5\pi}{6}+i\sin\frac{5\pi}{6}\right)$.

Exercise 2.3 · Computing a modulus without expanding. Compute the modulus of $(1+i)^6\left(\sqrt{3}-i\right)$, then that of $\dfrac{3+4i}{1-2i}$.

Exercise 2.4 · Sets of points. Determine and draw the set of points M with affix z such that:

1. $|z-2+i|=3$;
2. $|z-1|=|z+i|$;
3. $|\bar{z}+2i|=1$.

Exercise 2.5 · Nature of a triangle. Consider the points A, B, C with affixes $a=1+i$, $b=3+2i$ and $c=2+4i$. Compute AB, BC and AC, then determine the nature of the triangle ABC.

Exercise 2.6 · Numbers of modulus 1.

1. Prove that if $|z|=1$, then $z+\dfrac{1}{z}$ is real.
2. Let z and z' be two elements of $\mathbb{U}$ such that $zz'\neq -1$. Prove that $\dfrac{z+z'}{1+zz'}$ is real.

Exercise 2.7 · A sequence of complex numbers. Let $z_0=4$ and, for every natural number n, $z_{n+1}=\dfrac{1+i}{2}z_n$.

1. Compute z_1, z_2 and plot the corresponding points.
2. Prove that $(|z_n|)$ is a geometric sequence; give its common ratio and its general term.
3. Determine the limit of $|z_n|$. Interpret geometrically.
4. Determine the smallest integer n such that $|z_n|<0.01$.